

Table 1: Mean warm runtime (ms) on BPIC2017 per engine and access pattern. Darker cells indicate faster order-of-magnitude bands; bold marks the fastest band in each column.

Family	Model	P1: Control flow		P2: Queries		P3: OC-Perf		P4: BI/KPI	
		DFG	Variants	OCPQ	EKG	Sync	Sojourn	K1	K2
Relational	SQLite	23886.2	4537.3	28418.0	106623.3	38626.2	36081.9	2535.9	317.0
	DuckDB	59.5	114.5	47.3	38.5	251.5	99.0	47.0	26.9
Dataframe	Pandas	440.8	1148.2	148.3	172.4	3322.8	1813.4	99.9	51.4
	Polars	96.7	74.1	22.0	33.7	569.3	372.2	13.4	5.0
Graph	Kuzu	224.5	644.1	62.2	38.5	669.8	461.6	155.6	20.7
	Neo4j	1192.5	1949.1	555.2	406.1	9705.5	6484.9	2399.9	45.9
Custom	Rust4PM	3.4	53.0	26.7	29.0	65.7	49.4	7.5	0.4